

Jay Unadkat

MSc Information Technology

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Education

Dhirubhai Ambani University (formerly DA-IICT)

MSc Information Technology – CPI: 8.00/10

GLS University

BCA | CPI: 8.40/10

2025 – Present

Gandhinagar, Gujarat

2022 – 2025

Ahmedabad, Gujarat

Technical Skills

Languages: Python, C++, Java, JavaScript, SQL

AI/LLM: LangChain, LangGraph, RAG, AI Agents, Tool Calling, Vector Databases, OpenAI API, Hugging Face

Frameworks: React.js, Node.js, Express.js, Tailwind CSS, Streamlit

Databases: MongoDB, MySQL, PostgreSQL

Backend/Infra: Redis, BullMQ, REST APIs, JWT Authentication, Load Testing (Autocannon)

Developer Tools: Git, GitHub, Postman, VS Code, AWS

Coursework: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

Projects

VoyAgent - Multi-Agent AI Travel Planning System | Python, LangGraph, LangChain, FastAPI, Express.js, Next.js, MongoDB  

- Architected a **stateful multi-agent AI system** using **LangGraph**, orchestrating a Supervisor with specialized Planner, Flight, Hotel, Itinerary, Critic, and Response agents through graph-based workflows to generate personalized travel plans.
- Designed an **Agentic AI workflow** incorporating **Human-in-the-Loop (HITL)** checkpoints and a Critic Agent that validates budget, destination, and travel constraints, interrupting execution to collect user clarification before resuming the workflow.
- Engineered a highly concurrent execution pipeline using **fan-out/fan-in parallel orchestration**, enabling hotel, flight, and itinerary agents to execute simultaneously while maintaining shared conversational state and reducing end-to-end workflow latency.
- Leveraged **Structured Outputs**, **Pydantic schemas**, and **SerpAPI** to generate strongly typed travel recommendations grounded in real-time flight and hotel information, significantly reducing pricing and availability hallucinations.
- Built a production-ready full-stack platform using **Next.js**, **Express.js**, **FastAPI**, and **MongoDB** featuring JWT authentication, persistent multi-turn conversations, REST APIs, and modular AI workflow orchestration.

CodeStage - Online Code Evaluation Platform | React.js, Node.js, Express.js, MongoDB, Redis, BullMQ, Judge0 API  

- Architected and deployed a full-stack coding platform enabling automated multi-test-case evaluation, with a decoupled backend supporting concurrent submission processing.
- Engineered an asynchronous, event-driven execution pipeline using **Redis** as a message broker and **BullMQ** for distributed job queueing, decoupling submission intake from code execution via horizontally scalable worker processes.
- Validated pipeline reliability under **concurrent** load via custom load-testing, achieving a **100% success rate** with zero dropped requests across **1000+ simultaneous** submission requests.
- Integrated the **Judge0 API** for sandboxed, isolated code execution, capturing compile/runtime diagnostics, execution time, and memory usage to deliver granular test-case-level feedback.
- Implemented a real-time status-polling mechanism on the frontend to reflect **asynchronous** job-lifecycle transitions (Queued → Processing → Completed), eliminating unresponsive UI states during evaluation.

CampusOne - Unified Digital Campus Management Platform | React.js, Node.js, Express.js, MongoDB, JWT  

- Developed a **unified digital campus platform** that addresses the problem of students and faculty having to **switch between multiple platforms** such as Moodle, Google Classroom, coding portals, and university ERP systems by centralizing essential academic services into a single interface.
- Developed a **role-based university ERP platform** using the MERN stack, providing separate dashboards and access controls for **Students, Faculty, and Administrators** through a unified authentication system.
- Implemented secure **JWT-based authentication** with **bcrypt password hashing** and protected API routes, ensuring users can access only the resources and dashboards permitted by their assigned roles.
- Designed a scalable **RESTful backend** using Node.js and Express.js with modular controllers, middleware, routes, and Mongoose models, supporting centralized management of university users and academic resources.
- Built responsive role-specific interfaces using **React.js**, **Vite**, **Tailwind CSS**, and **DaisyUI**, with protected client-side routing and automatic dashboard redirection based on authenticated user roles.

Achievements

- Secured 1st Runner-Up at HackOut'25 (Synapse, DA-IICT) among 1000+ teams, leading the team during the hackathon.
- Secured 2nd Runner-Up in FaceOff, a competitive 1v1 DSA/CP contest conducted by MSTC (Microsoft Student Technical Club).
- Solved 500+ Data Structures and Algorithms problems across LeetCode, Codeforces, and GeeksforGeeks. LeetCode Max Rating: 1808.

Position of Responsibility

Student Placement Cell, DA-IICT
Member

Nov 2025 – Present

- Coordinated end-to-end placement activities and served as a liaison between students and recruiting companies, ensuring smooth communication and execution of pre-placement sessions.